

| Question  | Answer                                                                                                            | Marks     |
|-----------|-------------------------------------------------------------------------------------------------------------------|-----------|
| 2(a)      | same temperature (as each other)                                                                                  | <b>B1</b> |
|           | no net transfer of thermal energy (between them)                                                                  | <b>B1</b> |
| 2(b)(i)   | $E_1 = XL$                                                                                                        | <b>B1</b> |
| 2(b)(ii)  | $E_2 = Mc(t - \theta)$                                                                                            | <b>A1</b> |
| 2(b)(iii) | $E_3 = Xc\theta$                                                                                                  | <b>B1</b> |
| 2(c)      | $E_2 = E_1 + E_3$                                                                                                 | <b>C1</b> |
|           | $Mc(t - \theta) = XL + Xc\theta$<br><b>and</b><br>completion of algebra to reach $\theta = (Mct - XL) / c(M + X)$ | <b>A1</b> |

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